

Support Vector Machines (SVM)

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Support Vector Machine

- Support vector machine is a generalization of a simple classifier called maximal margin classifier
- However maximal margin classifier cannot be applied to most datasets since it requires linearly separable classes
- Support vector classifier is an extension of the maximal margin classifier that can be applied in a broader range of cases
- Support vector machine is a further extension of the support vector classifier to accommodate non-linear class boundaries
- Support vector machines are intended for the binary classification setting
- There are extensions of SVMs when there are more than two classes

What is a Hyper Plane?

- In a p -dimensional space, a hyperplane is a flat affine subspace of dimension $p-1$.
- In two dimensions, a hyperplane is a flat one-dimensional subspace or a line.
- In three dimensions, a hyperplane is a flat two-dimensional subspace or a plane.
- In $p > 3$ dimensions, it is hard to visualize a hyperplane but the notion of a $(p-1)$ -dimensional flat subspace still applies.

What is a Hyper Plane?

- In two dimensions, a hyperplane is defined by the equation,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

- In p dimensions, a hyperplane is defined by the equation,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

- Points in the data set should satisfy this equation.

What is a Hyper Plane?

- If X does not satisfy the equation,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0$$

- Then this tells us that X lies on one side of the hyperplane.
- On the other hand if,

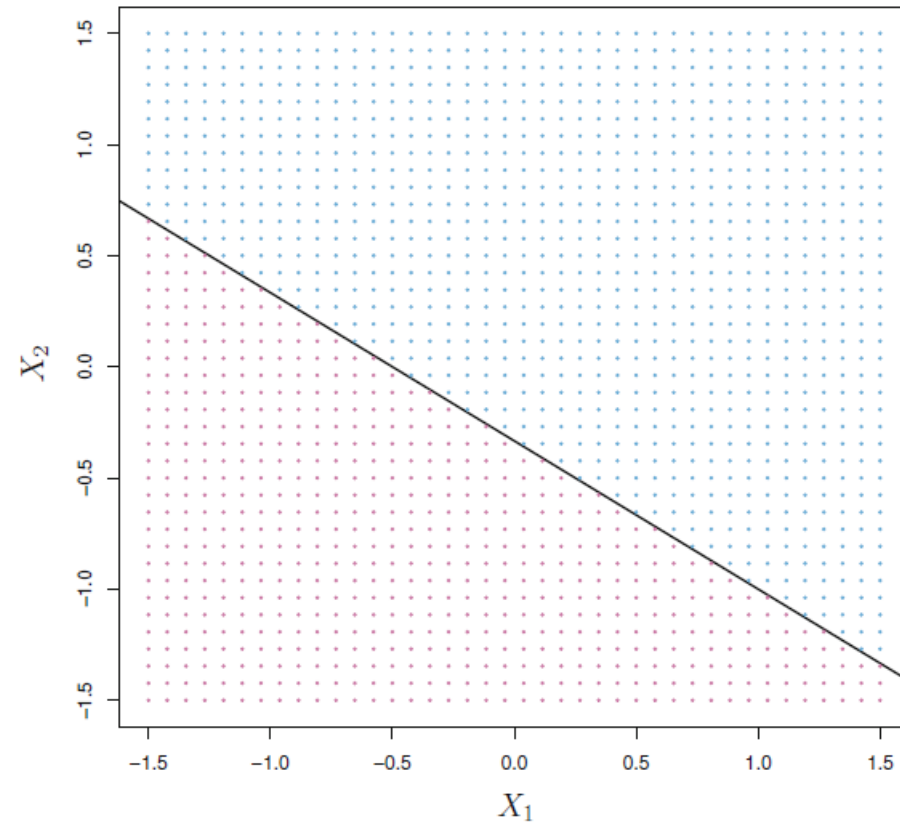
$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0$$

then X lies on the other side of the hyperplane.

- So the hyperplane divides p -dimensional space into two halves.

What is a Hyper Plane?

A hyper plane in a two-dimensional space can be shown as,



Classification using a separating Hyper Plane

Consider a p dimensional space with n observations,

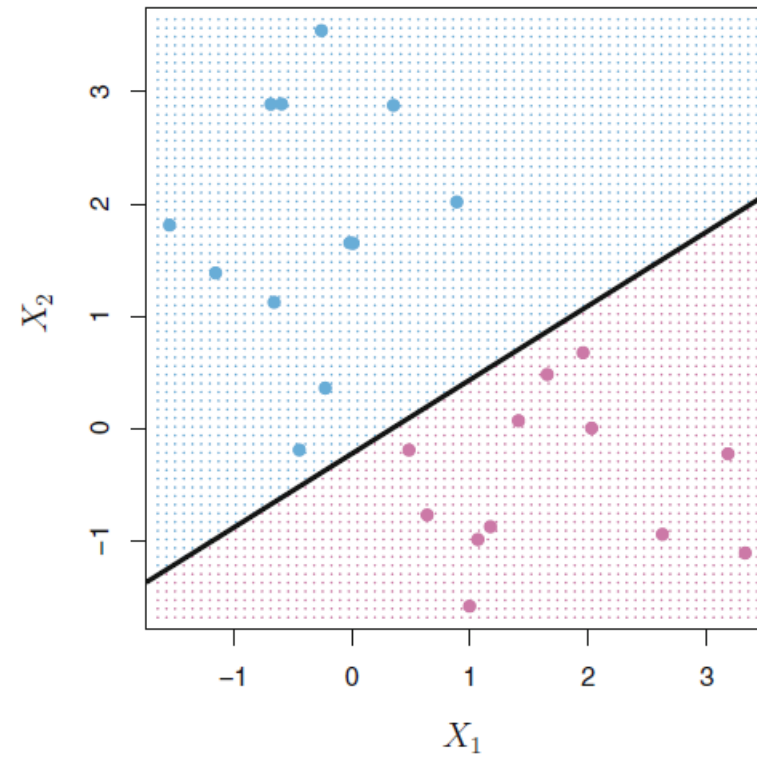
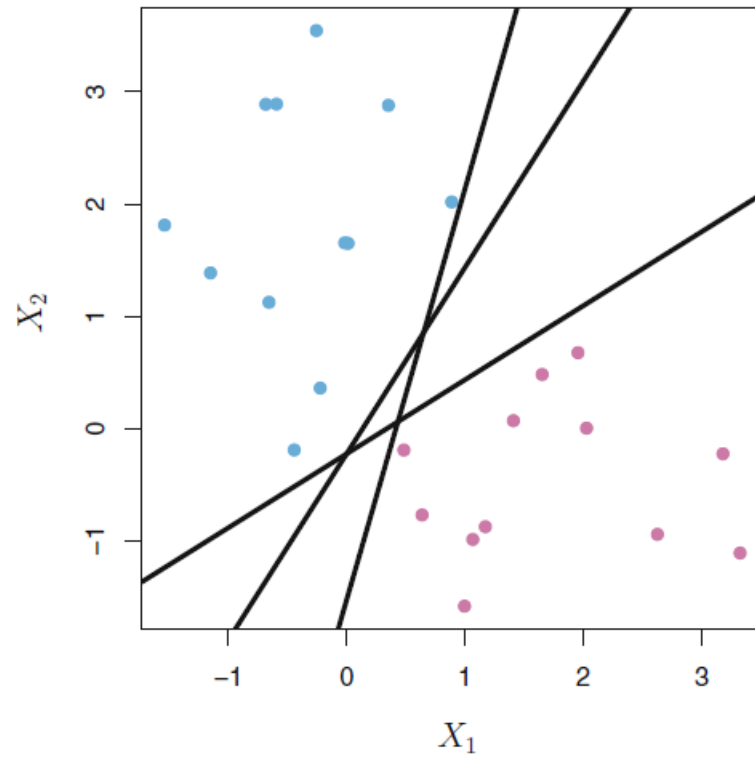
$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \dots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Consider that these observations fall into two classes -1 and 1.

$$y_1, \dots, y_n \in -1, 1$$

Classification using a separating Hyper Plane

Suppose that there is a hyperplane that perfectly separates the training observations to their classes.



Classification using a separating Hyper Plane

This hyper plane has the property that,

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1,$$

and

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1,$$

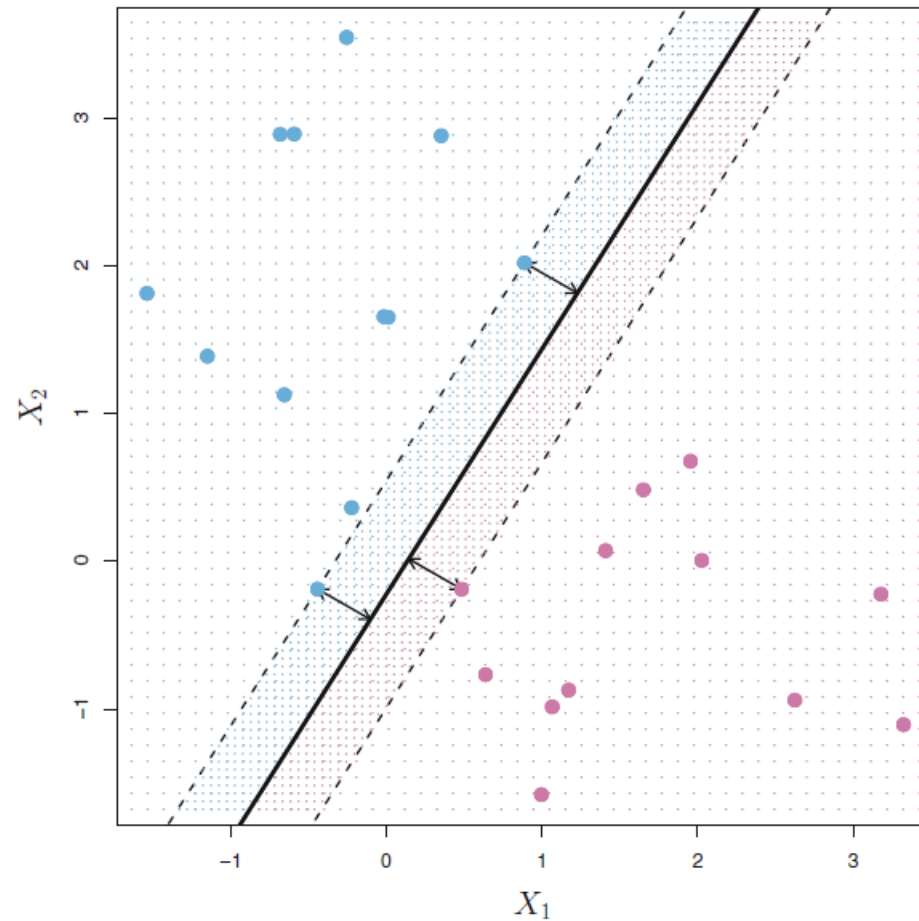
Equivalently a separating hyperplane has the property that,

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

for all $i = 1, \dots, n$

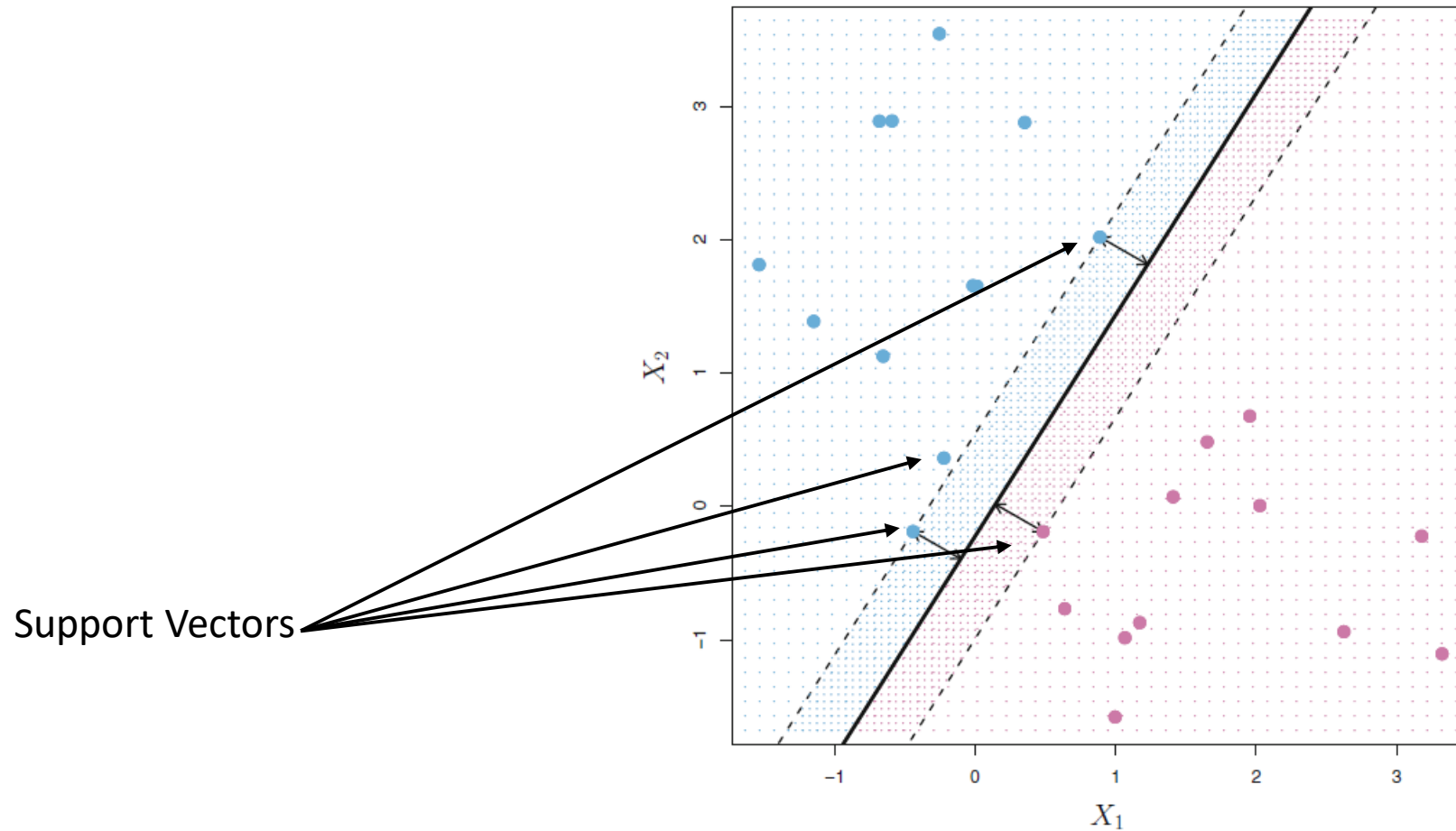
Maximal Margin Classifier

A natural choice is the maximal margin hyperplane (or optimal separating hyperplane) which is the separating hyperplane that is farthest from the training observations.



Maximal Margin Classifier

Points on the margin or close to the margin are called as Support Vectors.



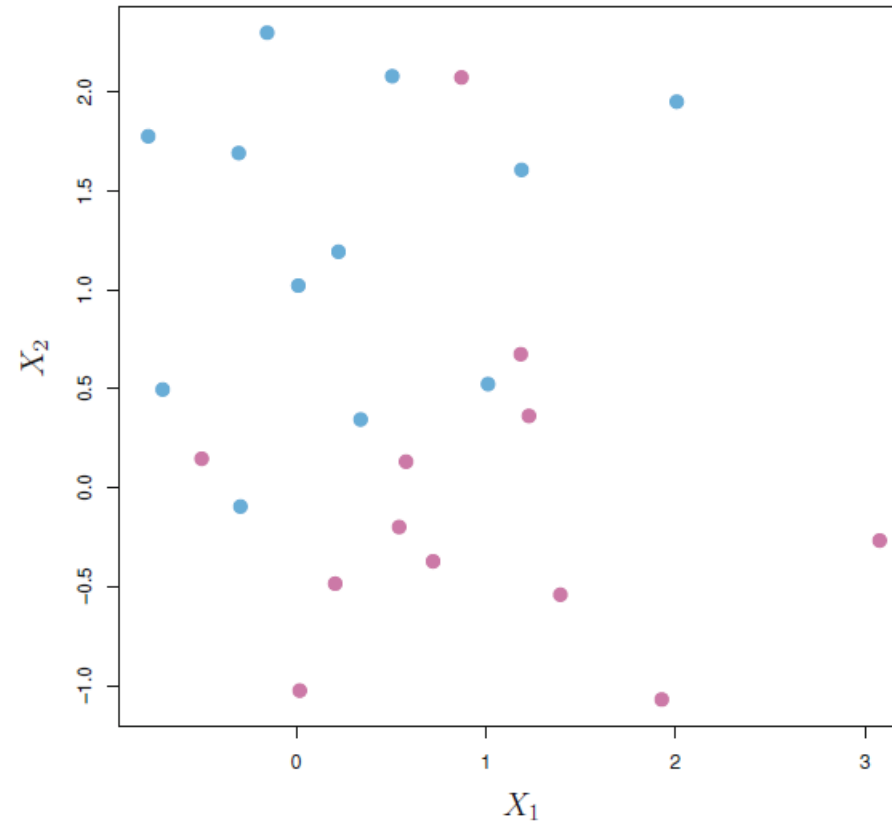
Maximal Margin Classifier

The maximal margin hyperplane is the solution to the optimization problem.

$$\begin{aligned} & \underset{\beta_0, \beta_1, \dots, \beta_p, M}{\text{maximize}} && M \\ & \text{subject to} && \sum_{j=1}^p \beta_j^2 = 1, \\ & && y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}) \geq M \quad \forall \quad i = 1, \dots, n \end{aligned}$$

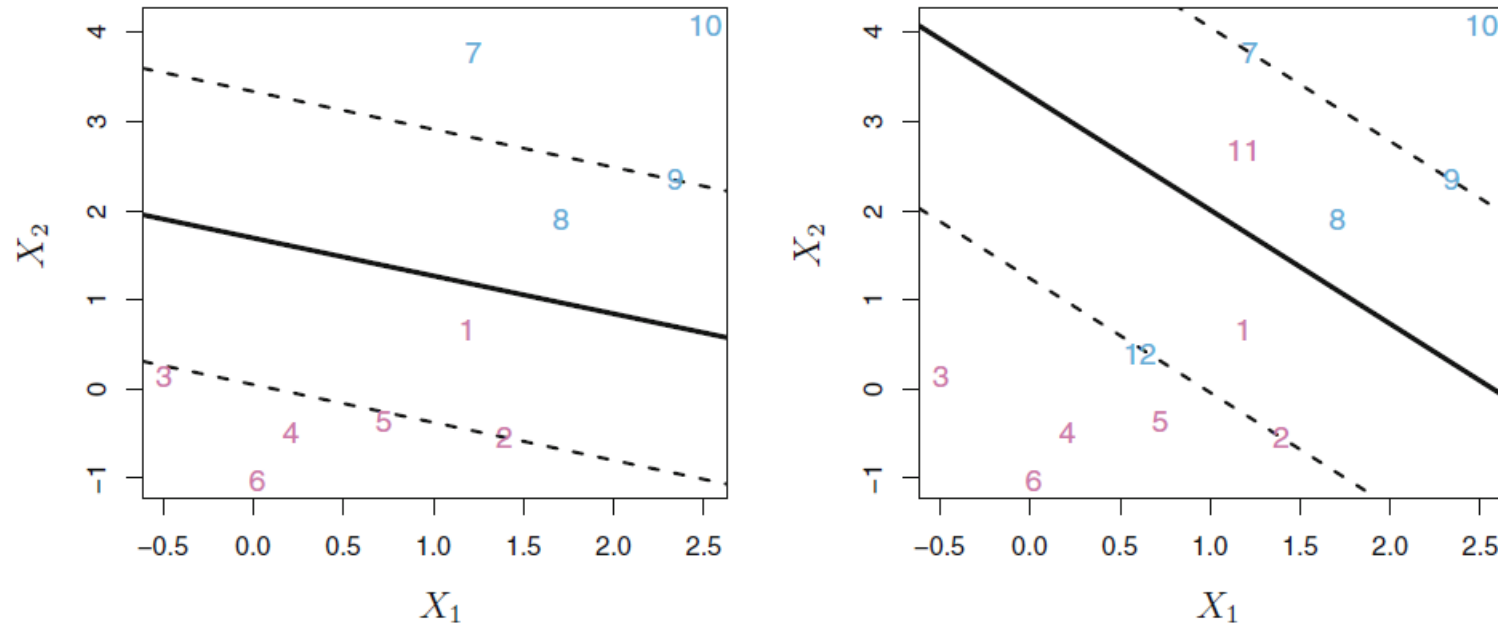
Non-Separable Data

Unfortunately, in many cases no separating hyperplane exists and there is no maximal margin classifier.



Support Vector Classifier (Soft Margin Classifier)

Consider the following example.



In the left panel, most of the observations are on the correct side of the margin. A small subset of the observations are on the wrong side. Observations on the wrong side of the hyperplane correspond to training observations that are misclassified (right panel).

Support Vector Classifier (Soft Margin Classifier)

Here the optimization problem is extended.

$$\begin{aligned} & \underset{\beta_0, \beta_1, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n, M}{\text{maximize}} && M \\ & \text{subject to} && \sum_{j=1}^p \beta_j^2 = 1, \\ & && y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}) \geq M(1 - \epsilon_i), \\ & && \epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq C \end{aligned}$$

Support Vector Classifier (Soft Margin Classifier)

Here $\epsilon_1, \dots, \epsilon_n$ are slack variables such that,

If $\epsilon_i = 0$, the i^{th} observation is on the correct side of the margin

If $\epsilon_i > 0$, the i^{th} observation is on the wrong side of the margin and it has violated the margin

If $\epsilon_i > 1$, it is on the wrong side of the hyperplane

C bounds the sum of the summation of slack variables.

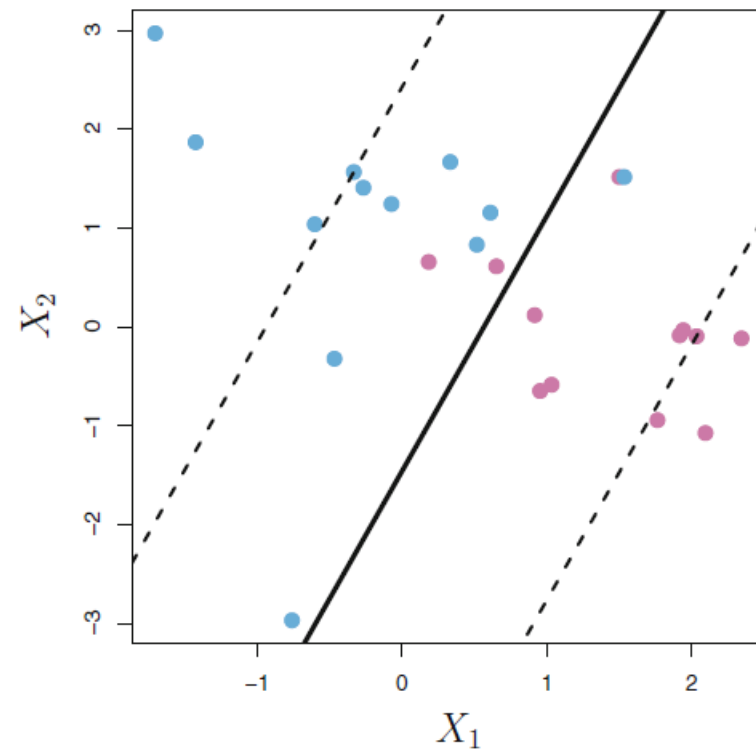
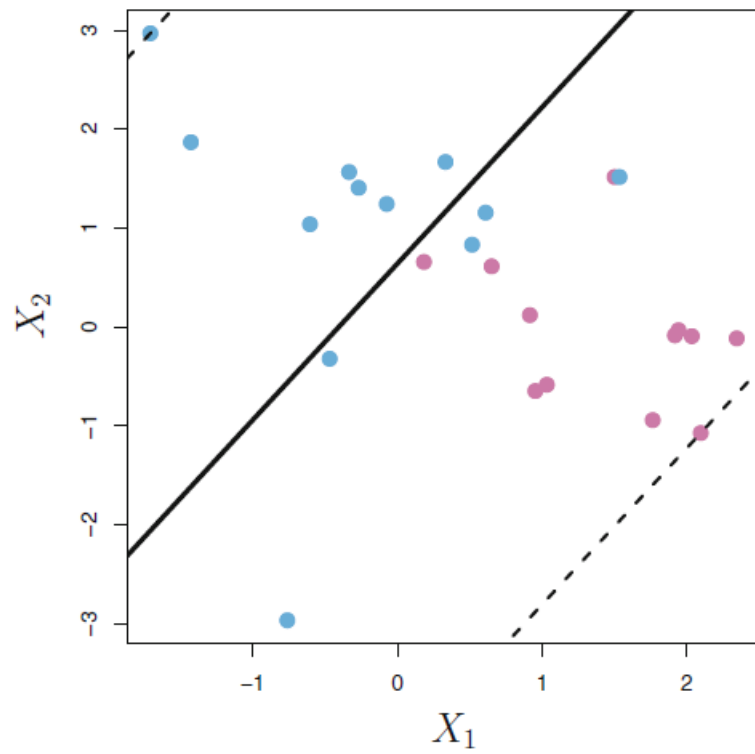
So it determines the number and severity of the violations to the margin (and to the hyperplane) that we tolerate.

C is a budget for the amount that the margin can be violated by the observations.

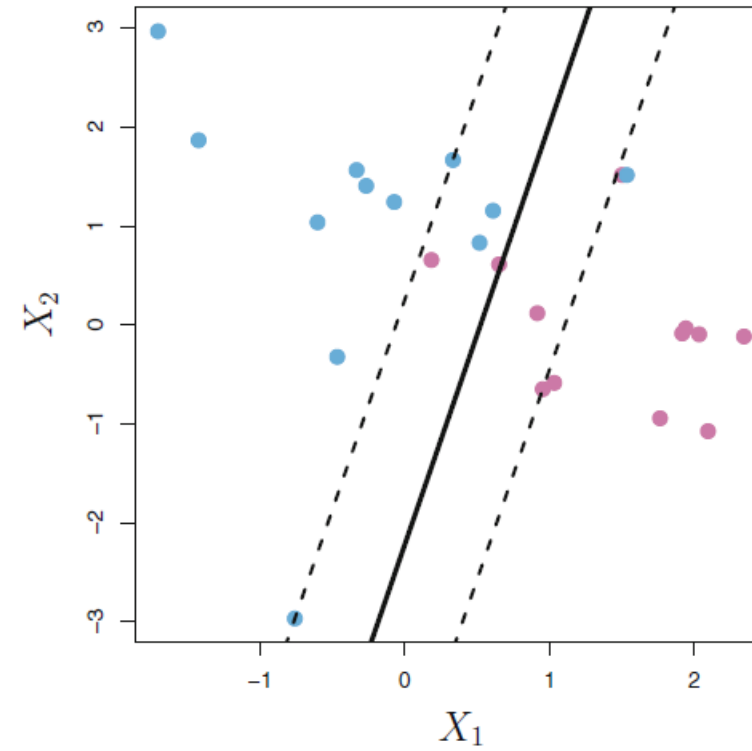
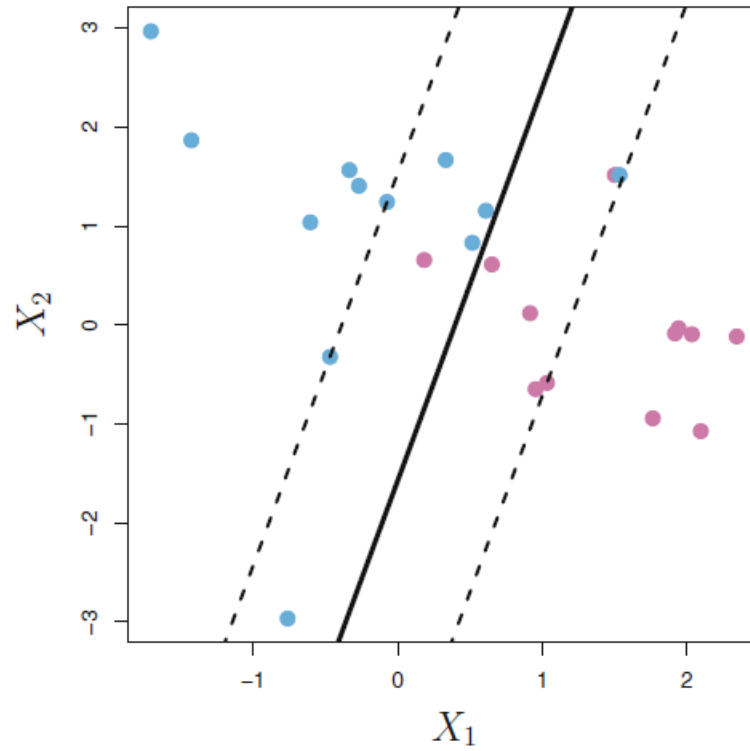
Support Vector Classifier (Soft Margin Classifier)

- If $C = 0$ then there is no budget for violations to the margin.
- For $C > 0$, no more than C observations can be on the wrong side of the hyperplane.
- As C increases, we become more tolerant of violations and the margin will widen.
- As C decreases, we become less tolerant of violations and the margin narrows.
- C is treated as a tuning parameter that is chosen via cross-validation.
- When C is small, we seek narrow margins that are rarely violated.
- A classifier that is highly fit to the data which may have low bias but high variance.
- When C is larger, the margin is wider and we allow more violations.
- A classifier that is potentially more biased but may have lower variance.
- Consider the following examples

Support Vector Classifier (Soft Margin Classifier)

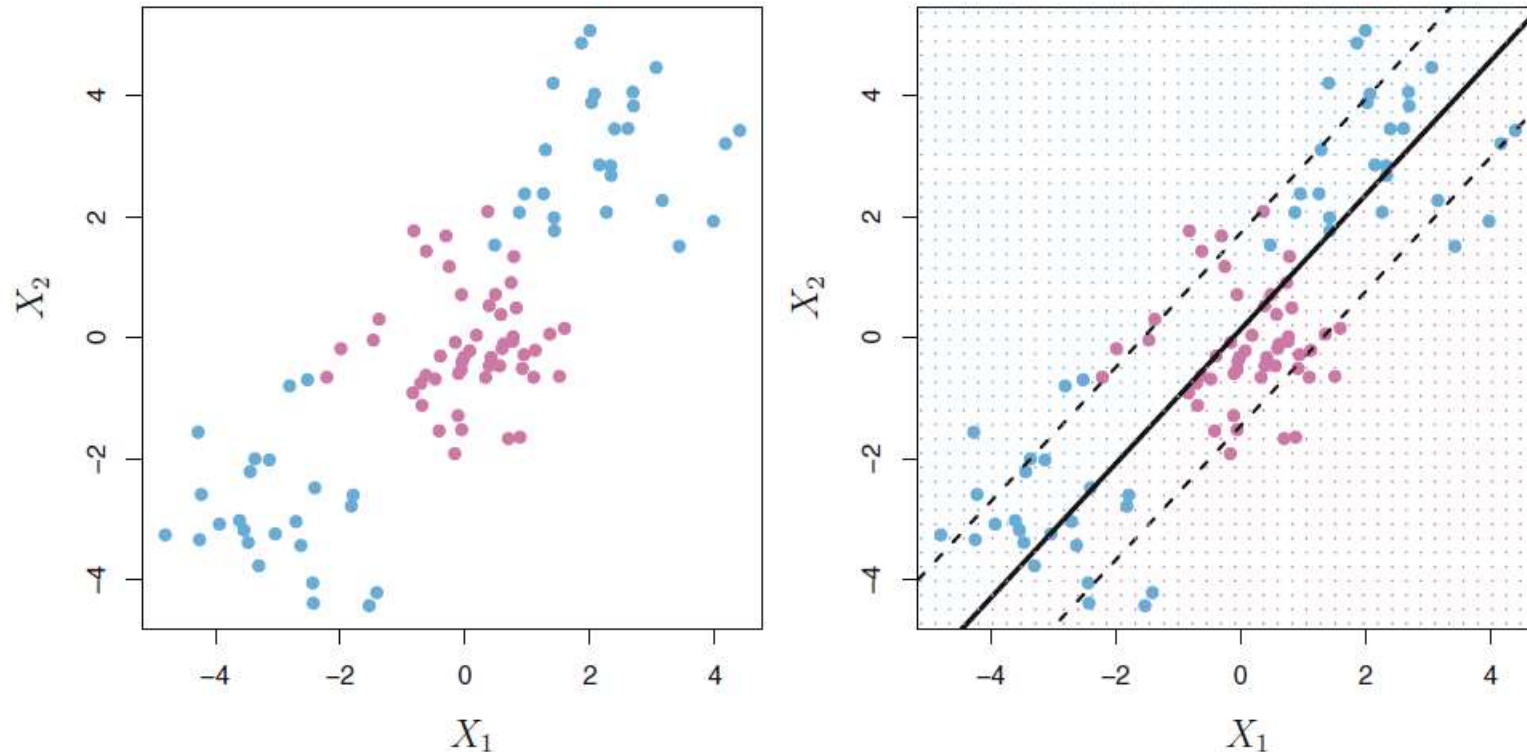


Support Vector Classifier (Soft Margin Classifier)



Classification with Non-linear Decision Boundaries

In practice we sometimes face non-linear class boundaries. A support vector classifier or any linear classifier will perform poorly here. The support vector classifier shown in the right panel below is useless here.



Classification with Non-linear Decision Boundaries

In linear regression, we enlarged the feature space using functions of the predictors to address non-linearity.

For instance, instead of fitting a support vector classifier using p features.

$$X_1, X_2, \dots, X_p$$

We can instead fit a classifier using $2p$ features

$$X_1, X_1^2, X_2, X_2^2, \dots, X_p, X_p^2$$

Classification with Non-linear Decision Boundaries

Then the optimization can be extended further,

$$\begin{aligned} & \underset{\beta_0, \beta_{11}, \beta_{12}, \dots, \beta_{p1}, \beta_{p2}, \epsilon_1, \dots, \epsilon_n, M}{\text{maximize}} \\ & \text{subject to } y_i \left(\beta_0 + \sum_{j=1}^p \beta_{j1} x_{ij} + \sum_{j=1}^p \beta_{j2} x_{ij}^2 \right) \geq M(1 - \epsilon_i), \\ & \sum_{i=1}^n \epsilon_i \leq C, \quad \epsilon_i \geq 0, \quad \sum_{j=1}^p \sum_{k=1}^2 \beta_{jk}^2 = 1 \end{aligned}$$

Support Vector Machines

There are many possible ways to enlarge the feature space. Support vector machine allows us to enlarge the feature space used by the support vector classifier in a computationally efficient way.

Support vector machine (SVM) is an extension of the support vector classifier.

The kernel approach is an efficient computational approach to implement this idea.

The main idea is to use **Kernel** functions in the classifier.

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

Support Vector Machines

There are several kernel functions available.

Linear Kernel

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \langle x, x_i \rangle$$

$$\langle x_i, x_{i'} \rangle = \sum_{j=1}^p x_{ij} x_{i'j}$$

Polynomial Kernel

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^p x_{ij} x_{i'j}\right)^d$$

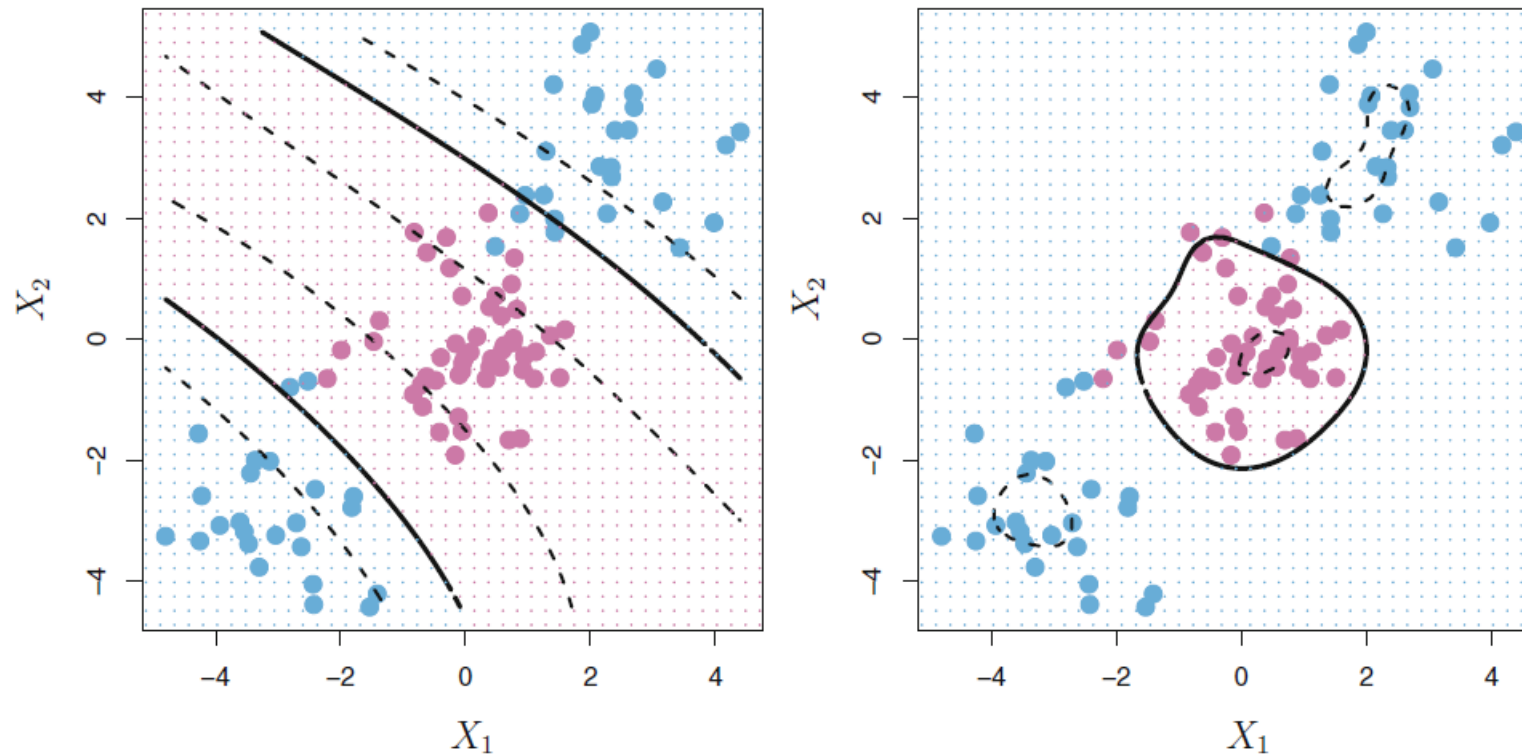
Radial Kernel

$$K(x_i, x_{i'}) = \exp\left(-\gamma \sum_{j=1}^p (x_{ij} - x_{i'j})^2\right)$$


$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

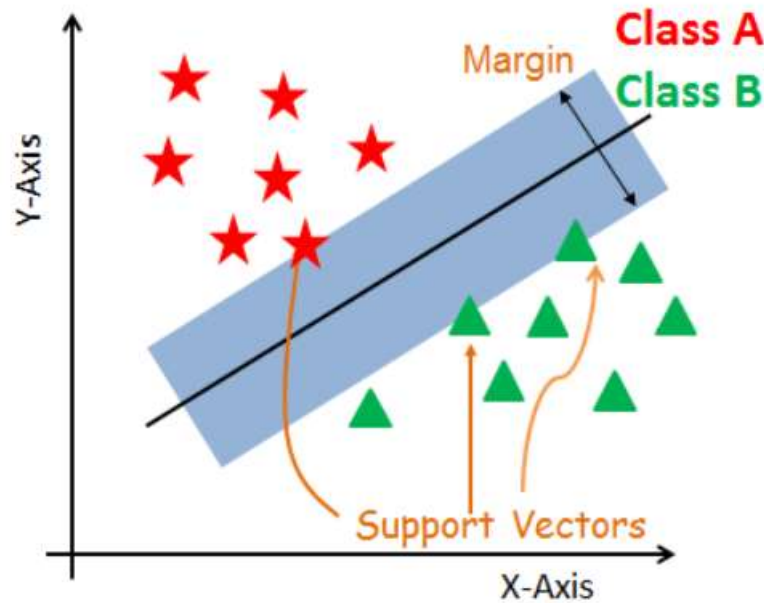
Support Vector Machines

The left panel shows an SVM with a polynomial kernel applied to the non-linear data. The right panel shows an example of an SVM with a radial kernel.



SVM – In Short

SVM constructs a hyperplane in multidimensional space to separate different classes. SVM generates optimal hyperplane in an iterative manner, which is used to minimize an error. The core idea of SVM is to find a maximum marginal hyperplane(MMH) that best divides the dataset into classes.

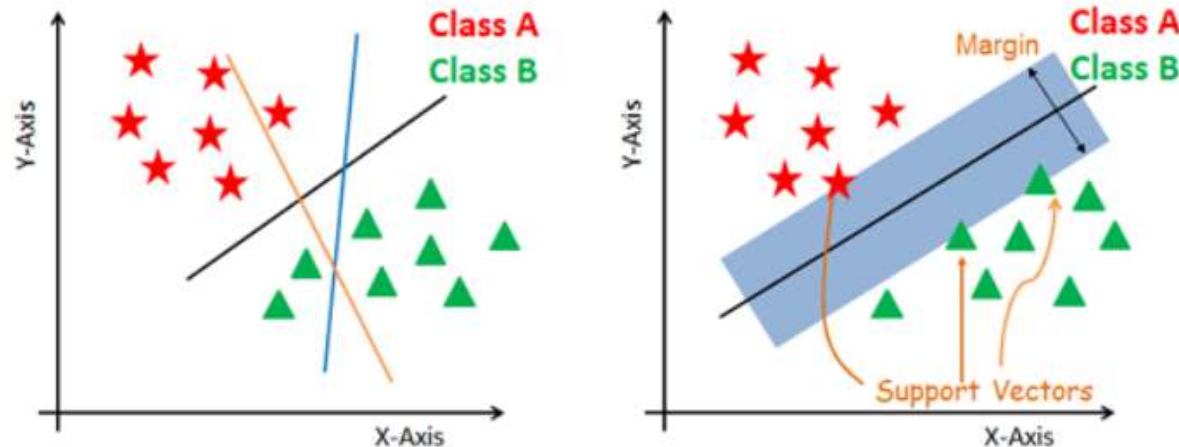


This is generally used for classification. But for regression, this can be extended.

SVM – In Short

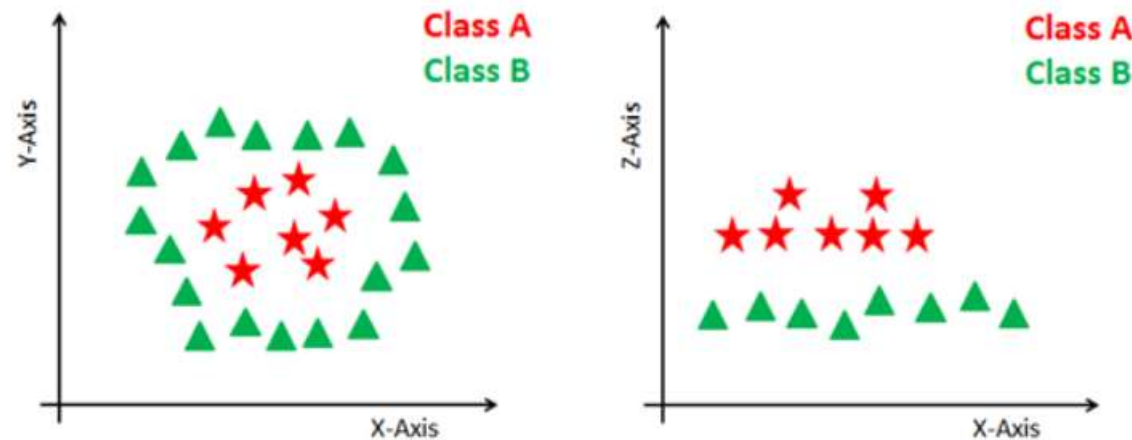
SVM searches for the maximum marginal hyperplane in the following steps:

1. Generate hyperplanes which segregates the classes in the best way. Left-hand side figure showing three hyperplanes black, blue and orange. Here, the blue and orange have higher classification error, but the black is separating the two classes correctly.
2. Select the right hyperplane with the maximum segregation from the either nearest data points as shown in the right-hand side figure.



SVM – In Short

Some problems can't be solved using linear hyperplane, as shown in the figure below. In such situation, SVM uses a kernel trick to transform the input space to a higher dimensional space as shown on the right.



Some popular kernels are,

- Linear Kernel
- Polynomial Kernel
- Radial Basis Function Kernel

Introduction to Support Vector Regression (SVR)

Support Vector Regression (SVR) uses the same principle as SVM, but for regression problems. Consider these two red lines as the decision boundary and the green line as the hyperplane. Our objective, when we are moving on with SVR, is to basically consider the points that are within the decision boundary line. Our best fit line is the hyperplane that has a maximum number of points.

